

Deep Learning Concepts-IT23D11

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Objectives:

- To understand the fundamental principles, theory and approaches for learning with deep neural networks.
- To learn the main variants of deep learning (such as convolutional and recurrent architectures), and their typical applications
- To understand the key concepts, issues and practices when training and modelling with deep architectures.
- To understand how deep learning fits within the context of other ML approaches
- To evaluate and optimize deep learning models

Why Neural Networks?

Applications

- Image Recognition
- Speech Recognition
- Medical Diagnosis
- Self-Driving Cars
- Fraud Detection
- Recommendation Systems

Example

- Input Image → Neural Network → Output
-  Cat Image → ANN → "Cat"

Introduction to Neural Networks

- Introduction
- Definition
- Working of Neural Networks
- Gradient Descent
- Stochastic Gradient Descent (SGD)
- Perceptron
- Multilayer Perceptron (MLP)
- Backpropagation

Introduction -**What is a Neural Network?**

- A **Neural Network (NN)** is a computational model inspired by the structure and functioning of the human brain. It consists of interconnected processing units called **neurons** that work together to learn patterns from data and make predictions or decisions.
- Neural Networks are a fundamental component of **Artificial Intelligence (AI)** and **Machine Learning (ML)** and form the basis of **Deep Learning (DL)**.

Human Brain vs Neural Network

Human Brain

Artificial Neural Network

Neurons

Artificial Neurons

Synapses

Weights

Signals

Input Data

Learning

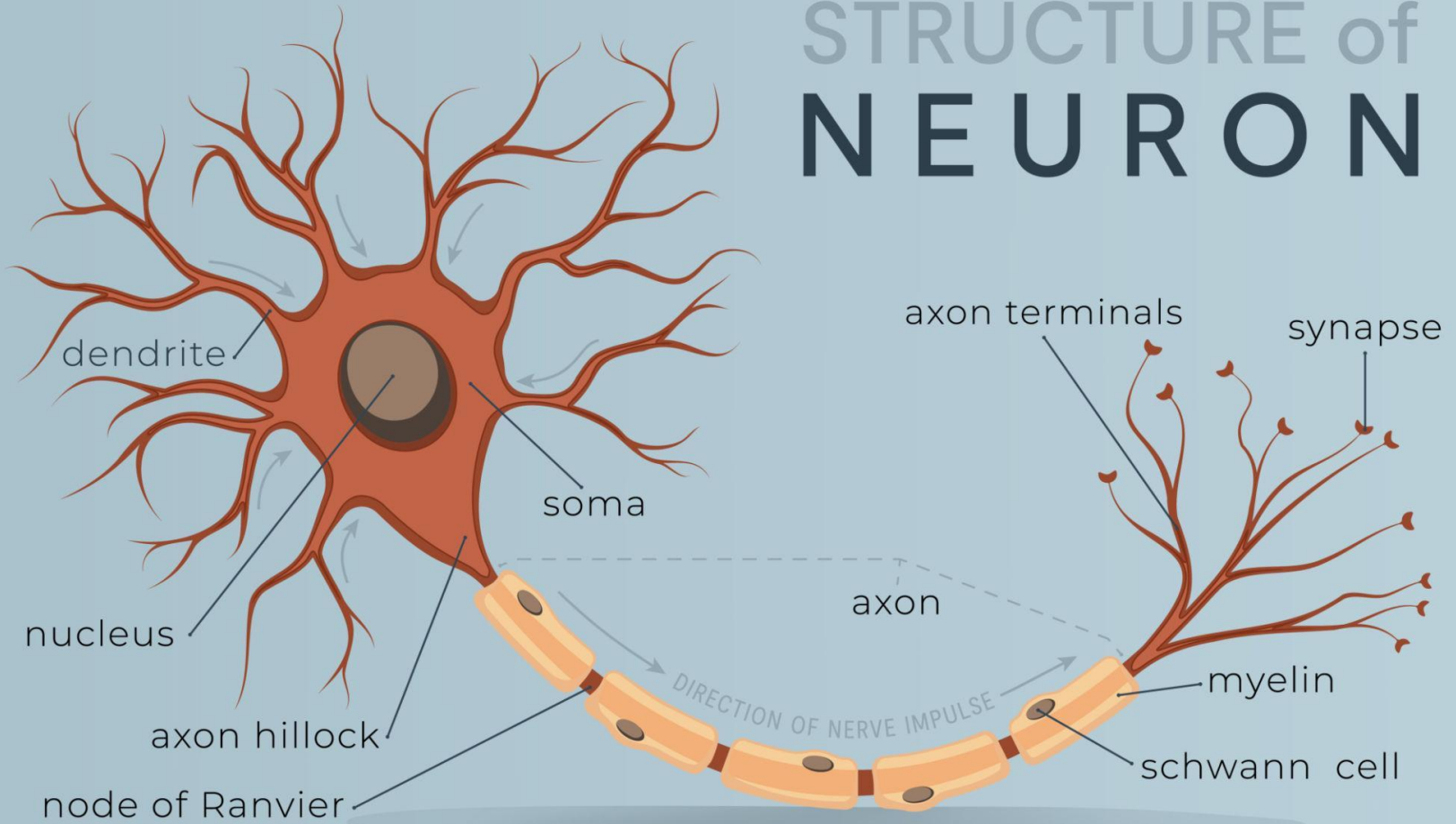
Training

Response

Output

Human Brain vs Neural Network

STRUCTURE of NEURON



Basic Structure of a Neural Network

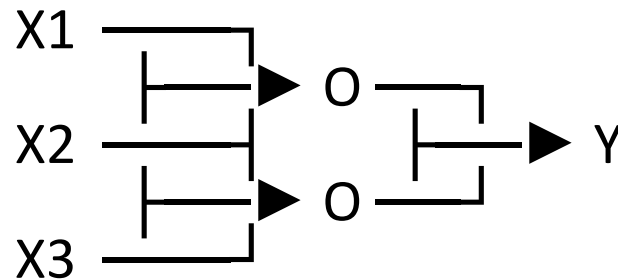
- A neural network typically consists of three types of layers:

1. Input Layer

- Receives data from the outside world.
- Example:
 - Age
 - Height
 - Weight

Cont..

- **2. Hidden Layer(s)**
- Processes information through mathematical computations.
- **3. Output Layer**
- Produces the final prediction or result.
- **Diagram**
- Input Layer Hidden Layer Output Layer

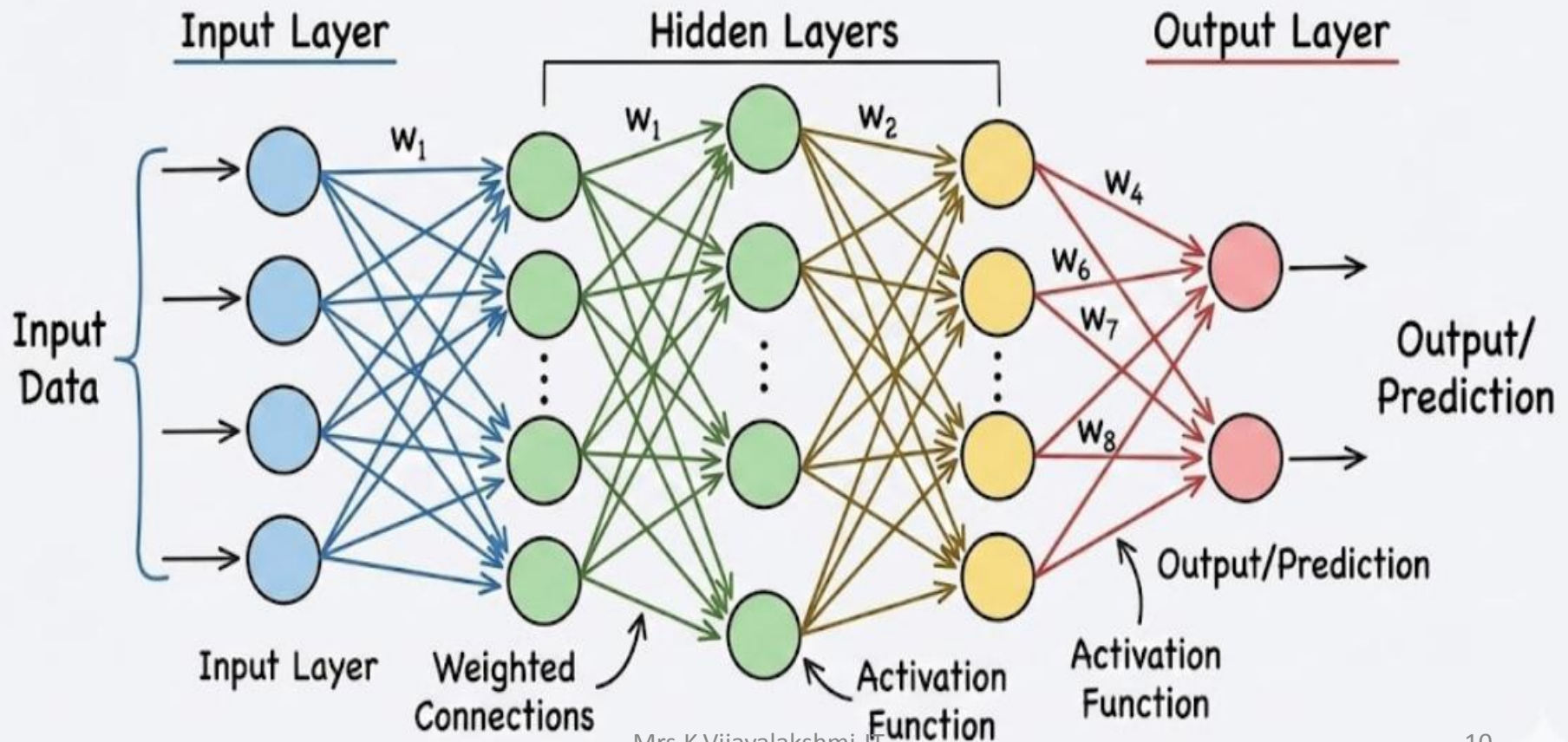


- Where:
- X1, X2, X3 = Inputs
- O = Neurons
- Y = Output

Cont..

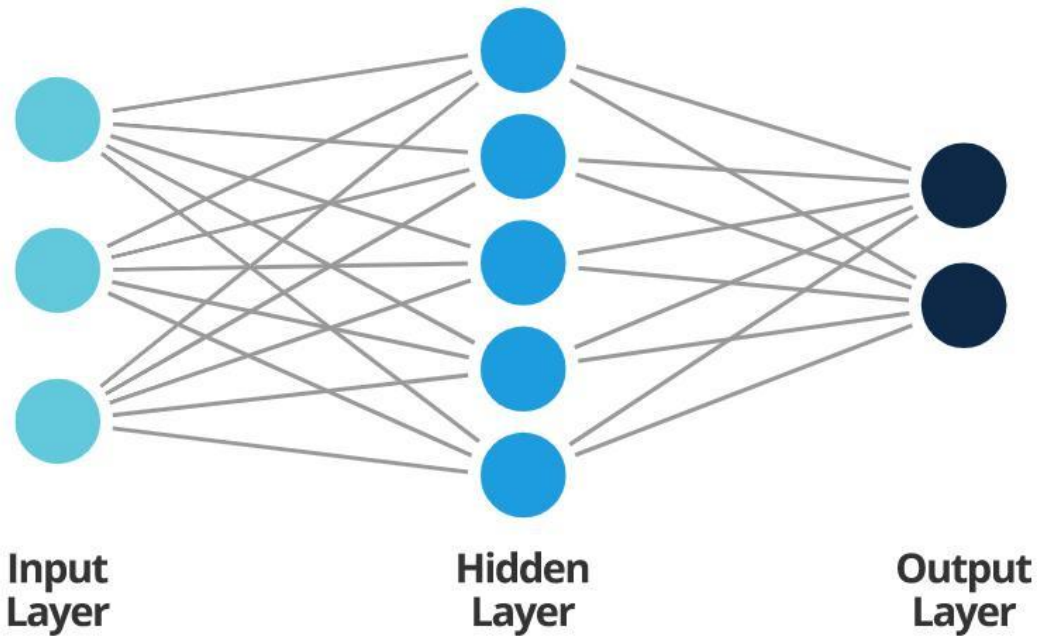
MACHINE LEARNING: ARTIFICIAL NEURAL NETWORK (ANN)

Data Processing & Pattern Recognition

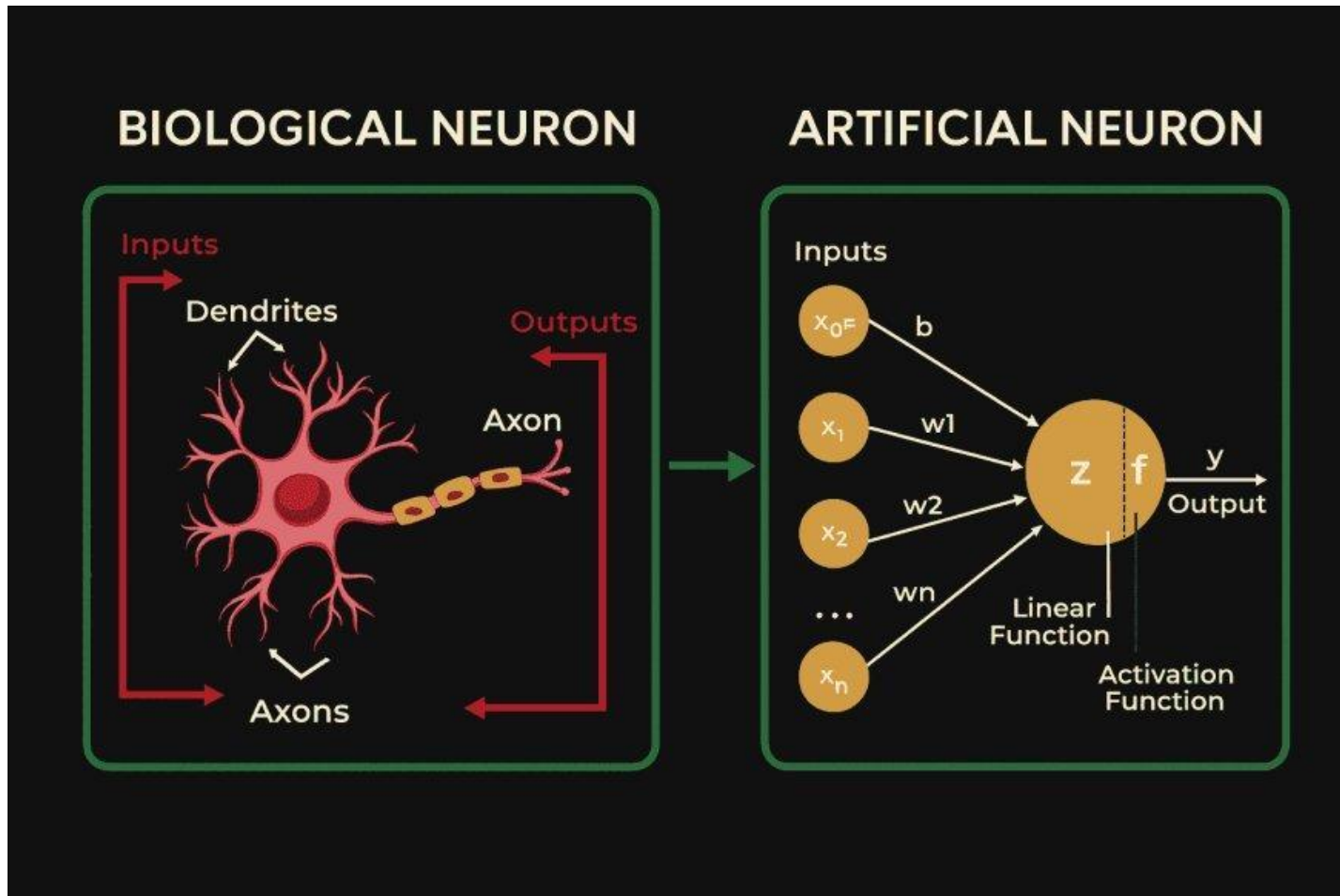


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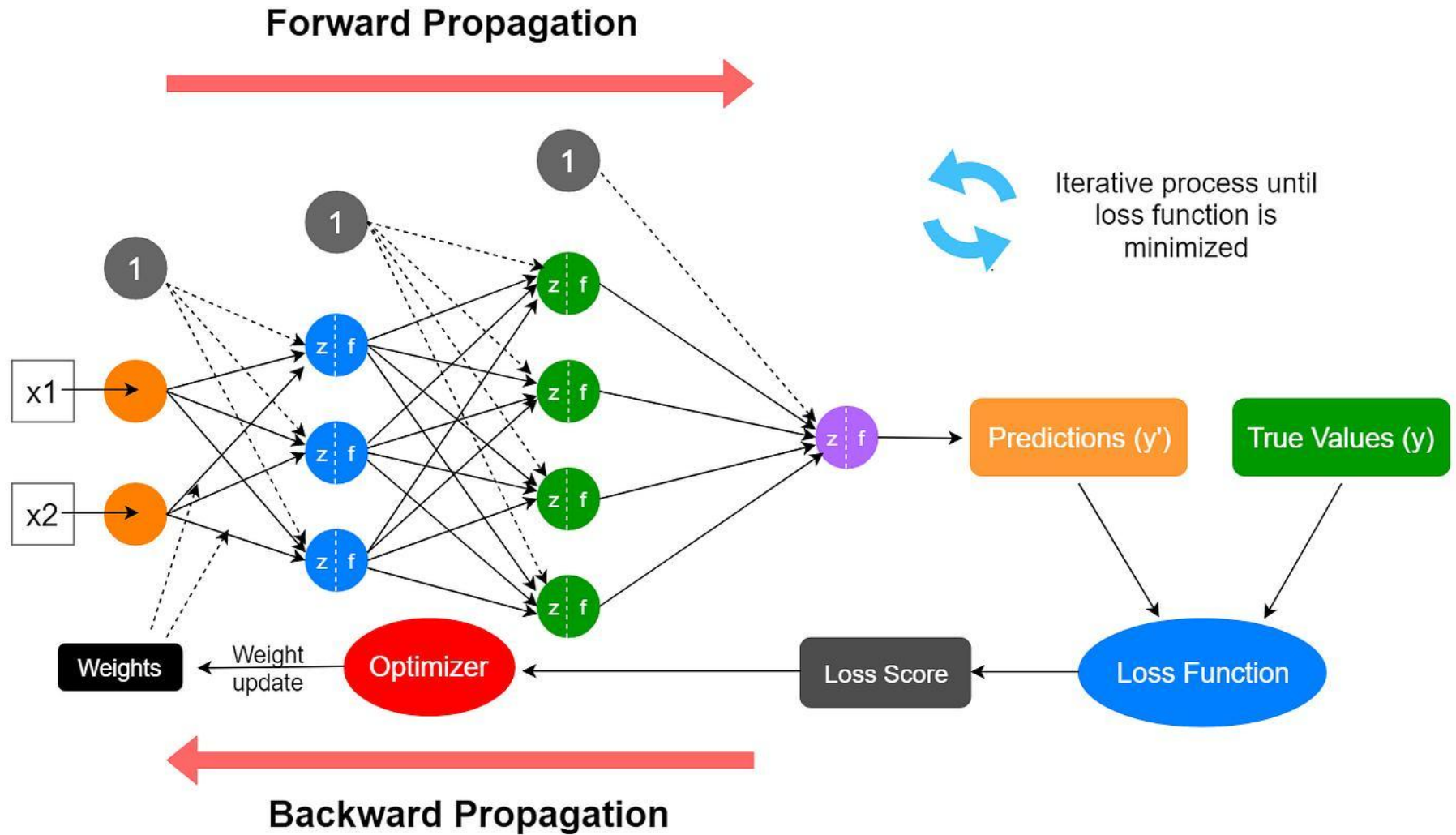
Artificial Neural Network Architecture



Cont..

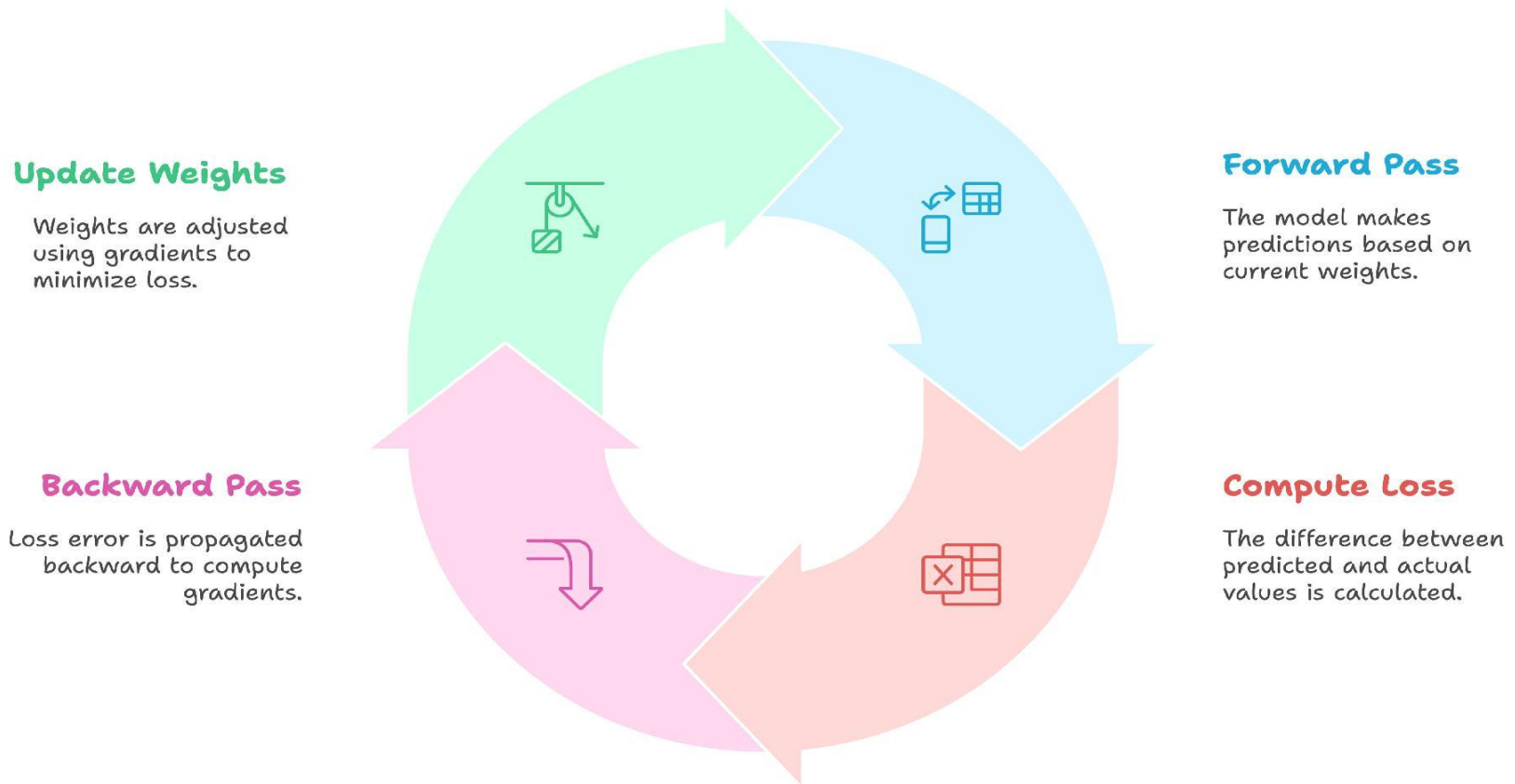


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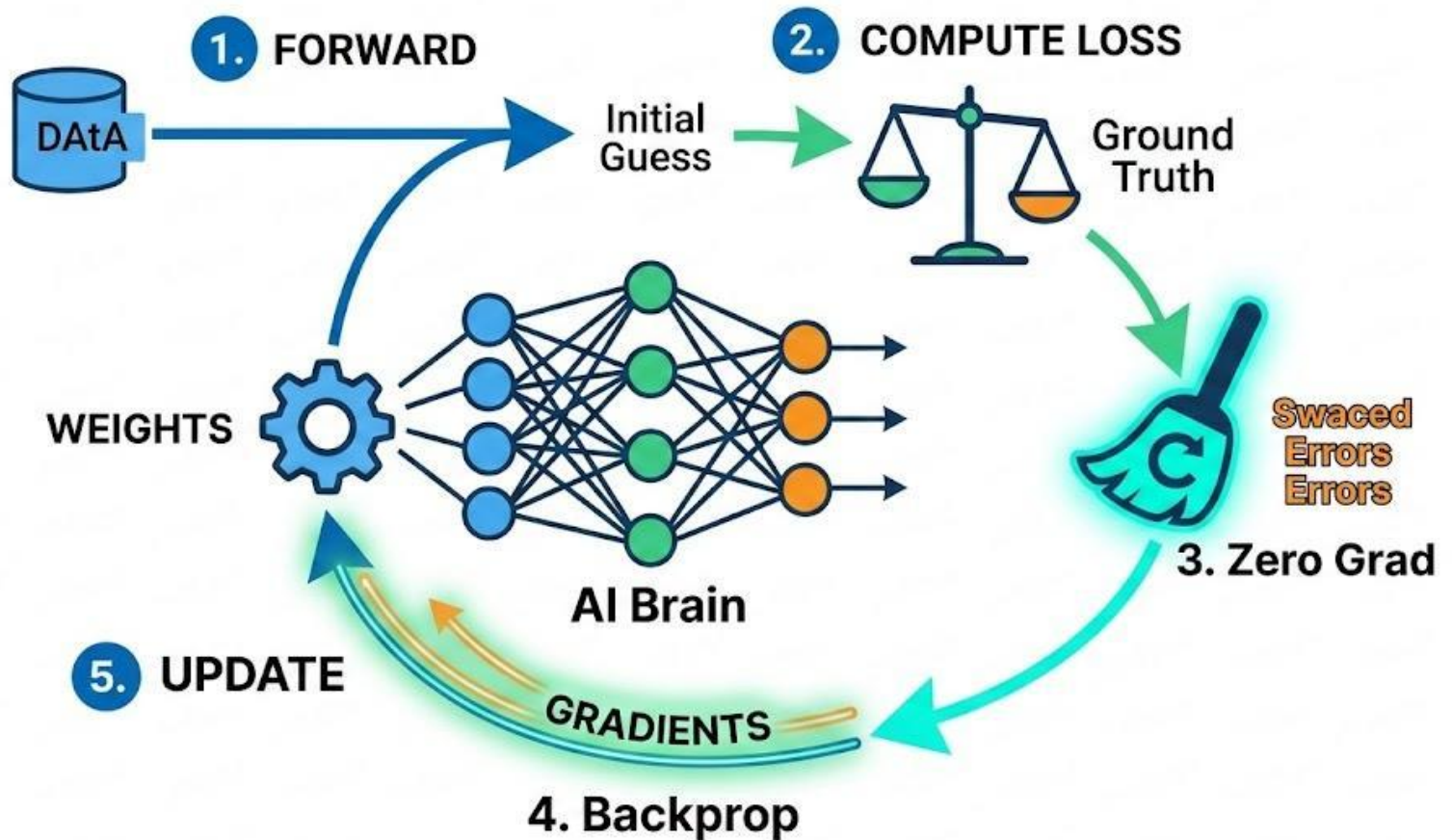


Cont...

Iterative Neural Network Training Cycle

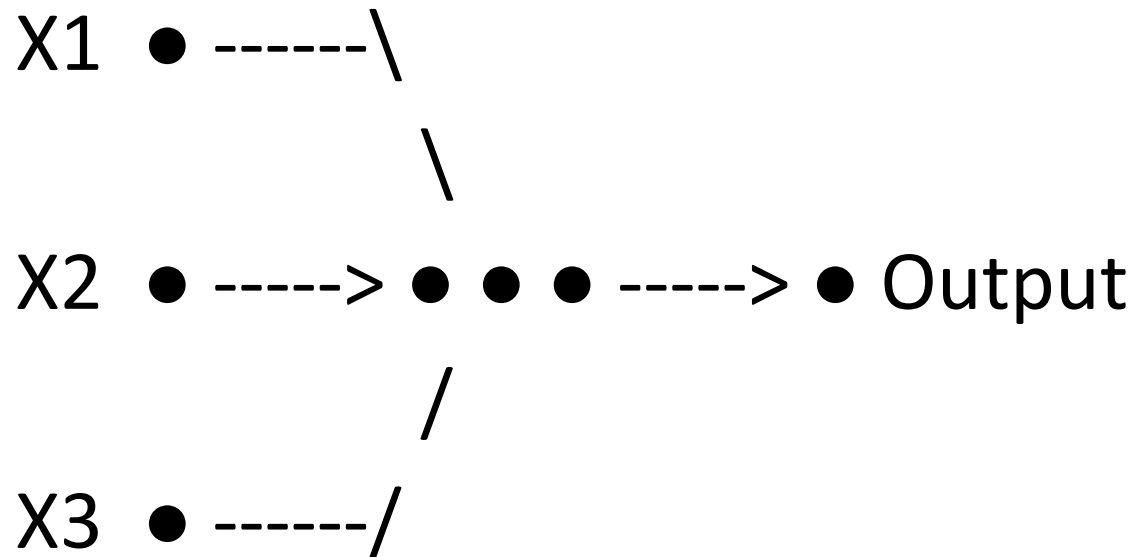


Cont..



Structure of Neural Network

- Input Layer Hidden Layer Output Layer



Artificial Neuron

- Input Layer receives data.
- Hidden Layer extracts features.
- Output Layer produces prediction.

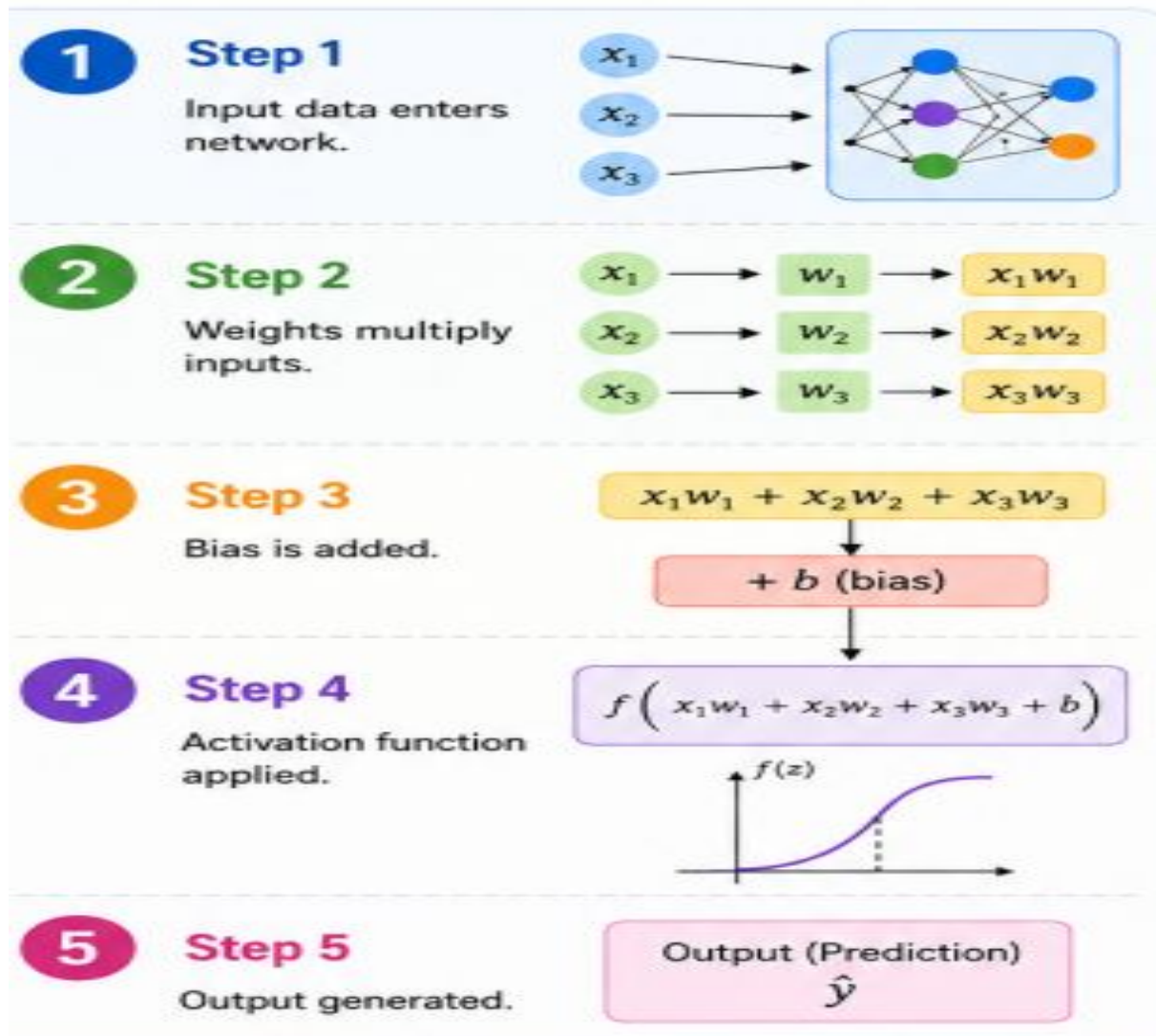
$$z = \sum(wx) + b, \text{ Output} = f(z)$$

- Where:
- x = Inputs
- w = Weights
- b = Bias
- f = Activation Function

Working of Neural Network

- Input → Weighted Sum → Activation → Prediction.

Cont..



Why Do We Need Activation Functions?

- **Example 1: College Exam Pass/Fail Prediction**

- Suppose an ANN receives the following student information:
- Study Hours = **8**
- Attendance = **90%**
- Assignment Marks = **85**

Without Activation Function

- The ANN calculates:
- Output = 28.75

Question:

- What does **28.75** mean?
- Pass or Fail?
- Good or Bad?
-  The result has **no practical meaning**.

Cont...

- **Example: Student Pass/Fail Prediction Using Sigmoid Activation Function**
- Suppose we want to predict whether a student will **pass or fail**.

Step 1: Input Data

Feature	Value
Study Hours	8 hours
Attendance	90%
Assignment Marks	85

These values are given to the **Input Layer** of the ANN.

Cont..

- Study Hours = 8
- Attendance = 90
- Assignment = 85



- Input Layer

Step 2: Multiply Inputs by Weights

- Each input has an importance (called a **weight**).
- Assume:

Input	Value	Weight
Study Hours	8	1.2
Attendance	90	0.2
Assignment	85	0.1

Cont..

- The ANN calculates:
- Study Hours contribution
- $8 \times 1.2 = 9.6$
- Attendance contribution
- $90 \times 0.2 = 18$
- Assignment contribution
- $85 \times 0.1 = 8.5$

Step 3: Add the Bias

- Bias = -7.35
- Now calculate the weighted sum:
- $Z = (9.6 + 18 + 8.5) - 7.35$
- $Z = 28.75$

Step 4: Why is 28.75 Not Useful?

- Suppose your teacher says:
- **"Your result is 28.75."**
- You would immediately ask:
- Did I pass?
- Did I fail?
- What does 28.75 mean?
- **Answer:** Nobody knows.
- It is only a mathematical value.

Step 5: Apply the Sigmoid Activation Function

- The Sigmoid formula is:
 - $f(x) = \frac{1}{1+e^{-x}}$
 - The ANN substitutes **28.75** into the formula.
 - Because **28.75 is a large positive number**, the output becomes approximately:
 - Sigmoid(28.75)
 - ↓
- 0.999999999...
- ≈ 1.00
- For teaching purposes, it is often rounded to **0.98** or **0.99** to represent a very high probability.

Step 6: Convert the Output into a Probability

The Sigmoid output always lies between **0 and 1**.

0 = Impossible

0.50 = 50% chance

1 = Certain

Our result:

0.98

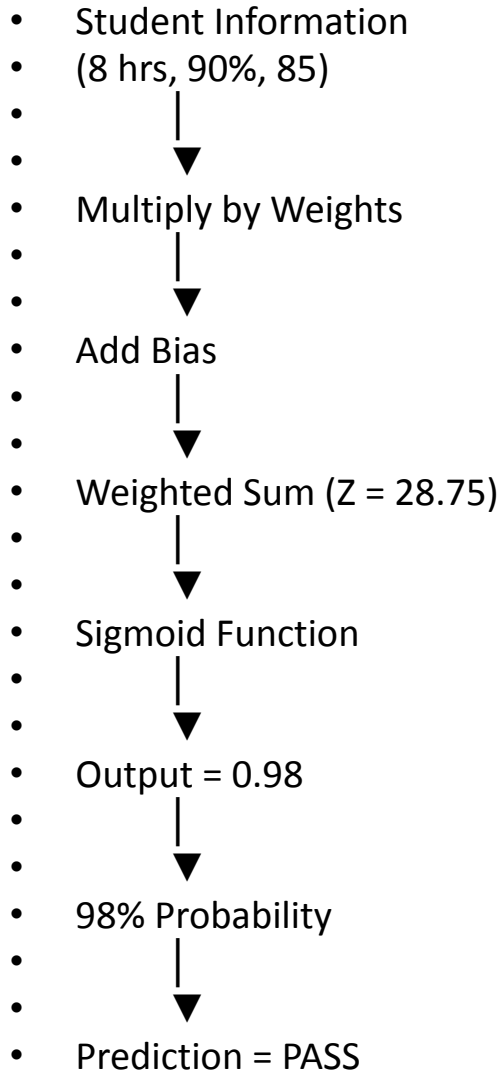
means

98% probability of passing

Step 7: Final Decision

- The ANN now compares the probability with a threshold. Usually,
- If Output > 0.5
- Student = PASS
- If Output < 0.5
- Student = FAIL
- Since
- $0.98 > 0.5$
- the ANN predicts:
- PASS

Complete Flow Diagram



Cont..

- the **Sigmoid activation function** converts the ANN's raw mathematical output into a **probability**, making it easy to interpret and use for decision-making.

Why Do We Need Activation Functions?

- Problem Without Activation Function
- Suppose a neural network performs only:
- $\text{Output} = WX + B$
- This is just a **linear equation**.
- Even if we add many hidden layers:
- Input $\rightarrow$ Hidden Layer 1 $\rightarrow$ Hidden Layer 2 $\rightarrow$ Output
- the network still behaves like a **single linear model**.

Result

-  Cannot learn complex patterns
-  Cannot recognize images, speech, or language
-  Cannot solve non-linear problems

Why Activation Functions Are Needed

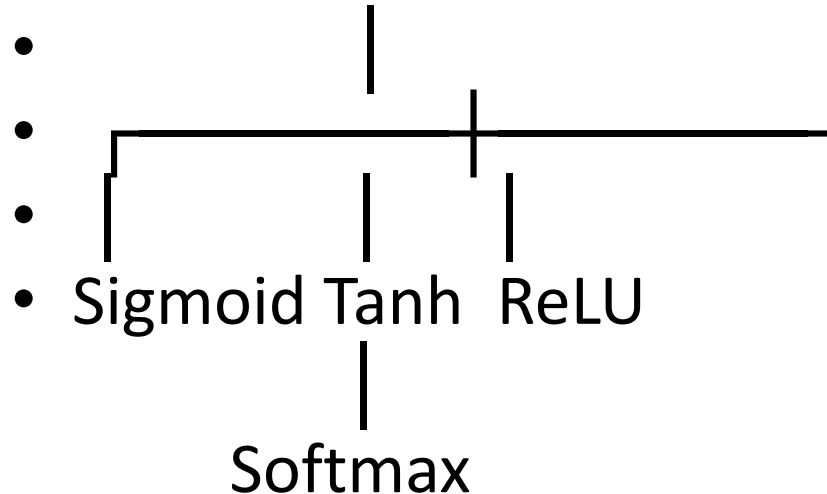
- Activation functions introduce **non-linearity** into the network.
- They help answer:
- "Should this neuron be activated or not?"

Types of Activation Functions

- Activation functions are mathematical functions that determine the output of a neuron and introduce **non-linearity** into neural networks.

Classification of Activation Functions

Activation Functions



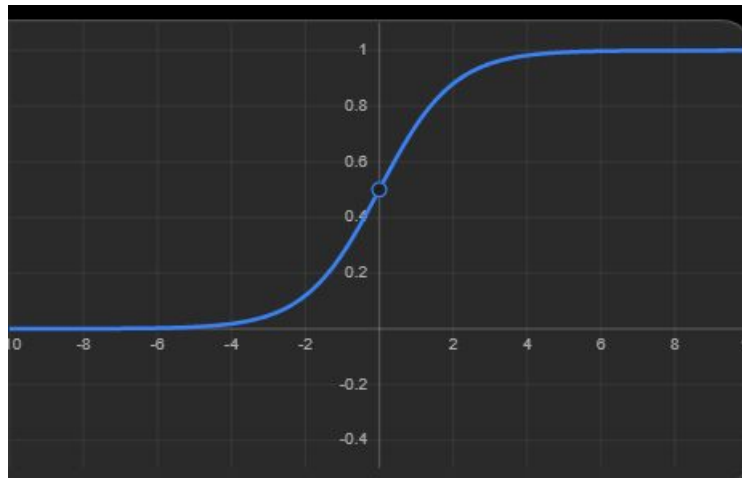
Benefits

- Learn complex relationships
-  Recognize patterns
-  Improve prediction accuracy
-  Make Deep Learning possible

1. Sigmoid Function

- **Formula**

- $$f(x) = \frac{1}{1+e^{-x}}$$



CONT..

- **Range**
 - **0 to 1**
 - **Characteristics**
 - S-shaped curve
 - Converts values into probabilities
 - **Advantages**
 - ✓ Smooth and differentiable
 - ✓ Suitable for binary classification
 - **Disadvantages**
 - ✗ Vanishing gradient problem
 - ✗ Slow convergence
 - **Applications**
- Binary classification
- Output layer of binary classifiers

CONT..

2. Hyperbolic Tangent (Tanh) Formula

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

CONT...

- **Range**
- **-1 to +1**
- **Characteristics**
- Zero-centered output
- Improved version of Sigmoid
- **Advantages**
- ✓ Better learning than Sigmoid
- ✓ Outputs can be positive or negative
- **Disadvantages**
- ✗ Still suffers from vanishing gradients
- **Applications**
- Hidden layers
- RNNs

3. ReLU (Rectified Linear Unit)

- **Formula** : $f(x) = \max(0, x)$
- **Range**
- **0 to ∞**
- **Characteristics**
- If $x < 0 \rightarrow \text{Output} = 0$
If $x \geq 0 \rightarrow \text{Output} = x$
- **Advantages**
- ✓ Fast computation
- ✓ Reduces vanishing gradients
- ✓ Most commonly used activation function
- **Disadvantages**
- ✗ Dead neuron problem
- **Applications**
- CNNs
- Deep Neural Networks

4. Softmax Function

- **Formula** : $Softmax(x_i) = \frac{e^{x_i}}{\sum e^{x_j}}$
- **Range**
- **0 to 1 (probabilities)**
- **Characteristics**
- Converts outputs into probability distribution
- Sum of all probabilities = 1
- **Example**
- Cat = 0.80
Dog = 0.15
Bird = 0.05

CONT..

- **Advantages**
- ✓ Excellent for multi-class classification
- ✓ Produces probability outputs
- **Disadvantages**
- ✗ Computationally expensive for many classes
- **Applications**
- Image classification
- Multi-class prediction

Which Activation Function Should Be Used?

Layer	Recommended Function
Hidden Layers	ReLU
Binary Classification Output	Sigmoid
Multi-Class Classification Output	Softmax
RNN/LSTM Hidden Layers	Tanh

Real-Life Example

Student Pass/Fail Prediction

- Input:
- Study Hours = 8
- Attendance = 90%

Without activation function:

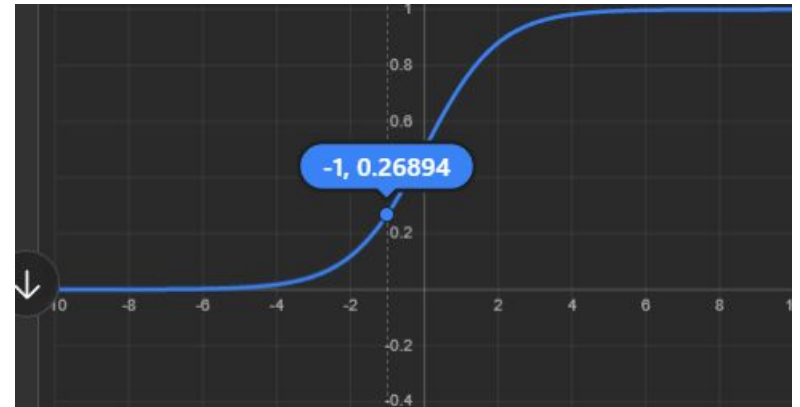
- Output = 25

What does 25 mean?

- No clear interpretation.

Example 1: Sigmoid Function

- Formula: $f(x) = 1/(1+e^{-x})$



Input (x)

-5

0

5

Output

0.006

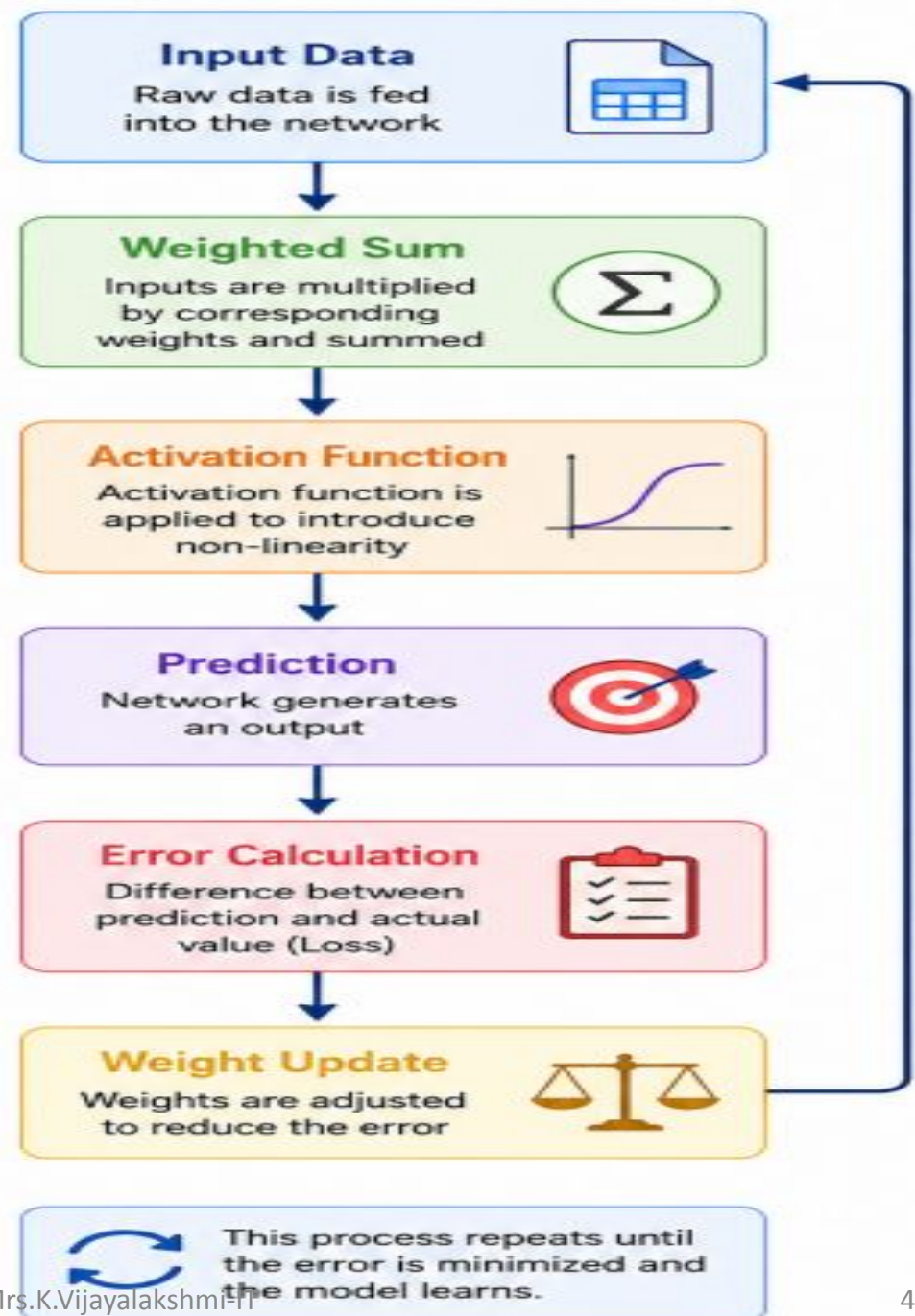
0.5

0.993

Use

- Binary Classification
- Yes/No Prediction
- Pass/Fail

- Cont..



Cont..

EXAMPLE

Inputs:

-  $x_1 = 0.6$
-  $x_2 = 0.3$
-  $x_3 = 0.9$

Weights:

- $w_1 = 0.5$
- $w_2 = -0.4$
- $w_3 = 0.8$

Bias:

$$b = 0.2$$

$$\begin{aligned}\text{Weighted Sum} &= x_1 w_1 + x_2 w_2 + x_3 w_3 + b \\ &= (0.6)(0.5) + (0.3)(-0.4) + (0.9)(0.8) + 0.2 \\ &= 0.3 - 0.12 + 0.72 + 0.2 \\ &= 1.10\end{aligned}$$

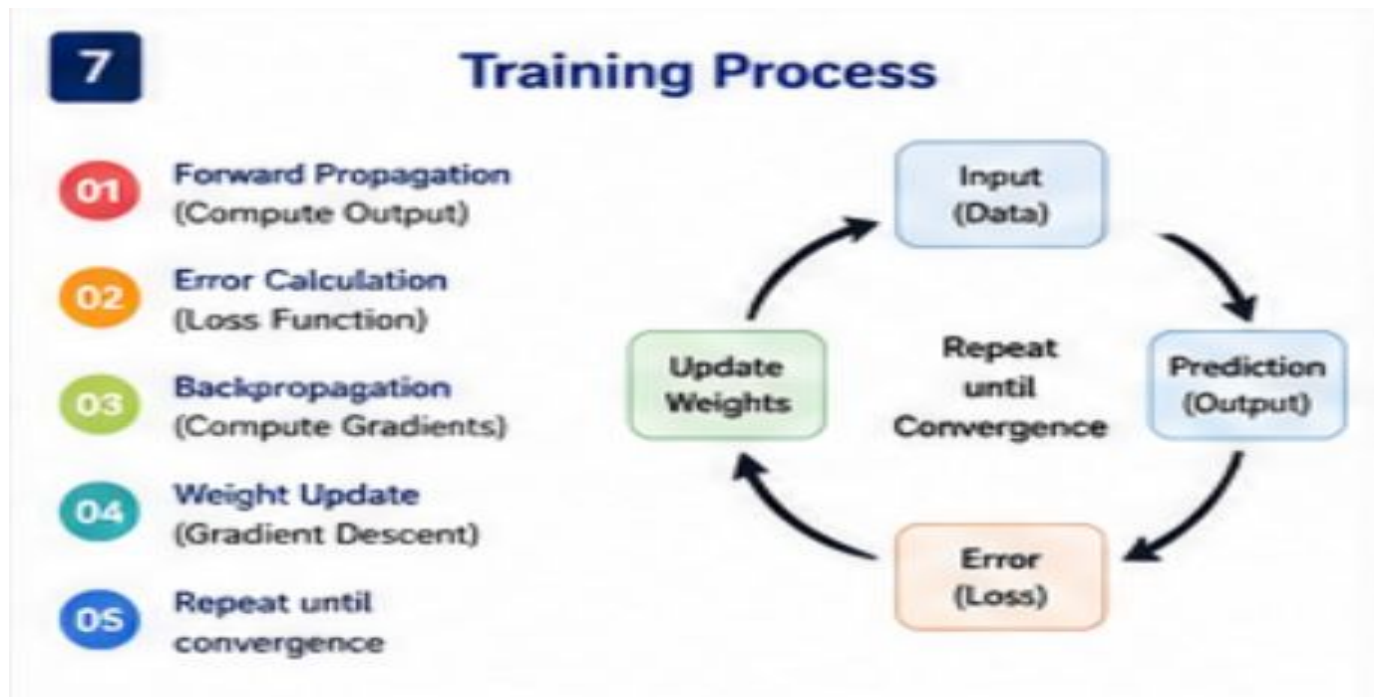
$$\text{Activation (Sigmoid): } f(z) = \frac{1}{1 + e^{-z}}$$

$$\text{Output (Prediction)} = f(1.10) \approx 0.75$$

So, the predicted output is **0.75**

Training Process

- Forward Propagation, Error Calculation, Backpropagation, Weight Update



Gradient Descent

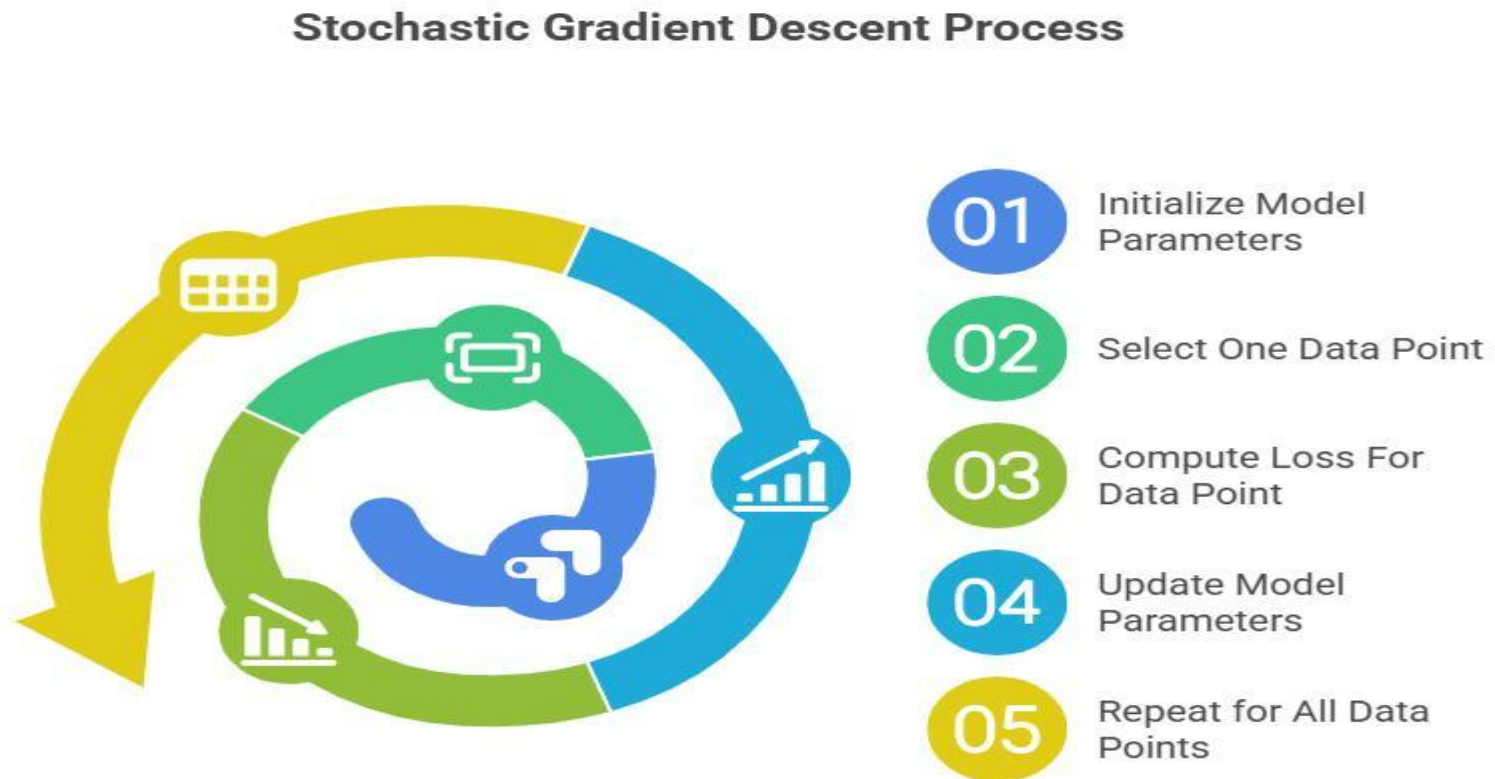
- Optimization algorithm used to minimize loss.

Definition

- **Gradient Descent (GD)** is an optimization algorithm used in Artificial Neural Networks (ANNs) and Deep Learning to **minimize the error (loss)** by **automatically updating the weights and biases during training.**

Stochastic Gradient Descent

- Updates weights using one training sample at a time.



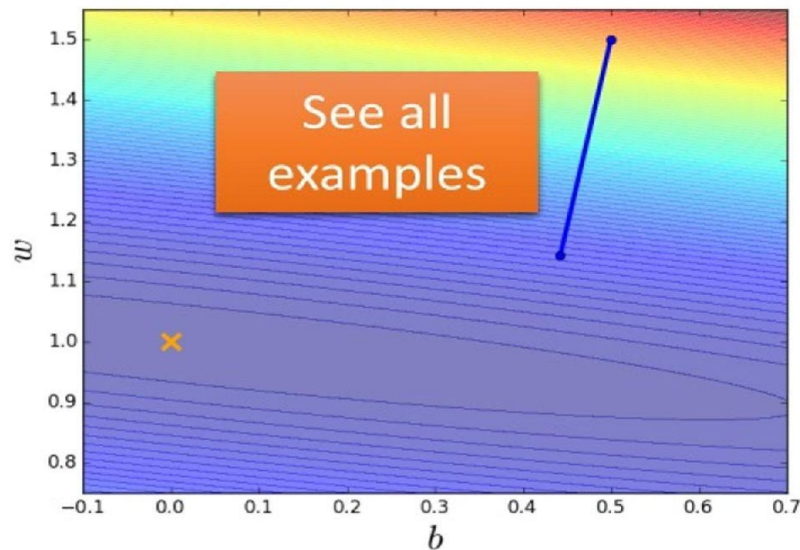
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- **In Gradient Descent:**
- The goal of Gradient Descent is to **minimize the loss (error)**.
- If the **learning rate is too high**, the algorithm may overshoot the minimum point.
- Instead of reducing the error, the **loss keeps increasing**. This is called **divergence**.

Stochastic Gradient Descent

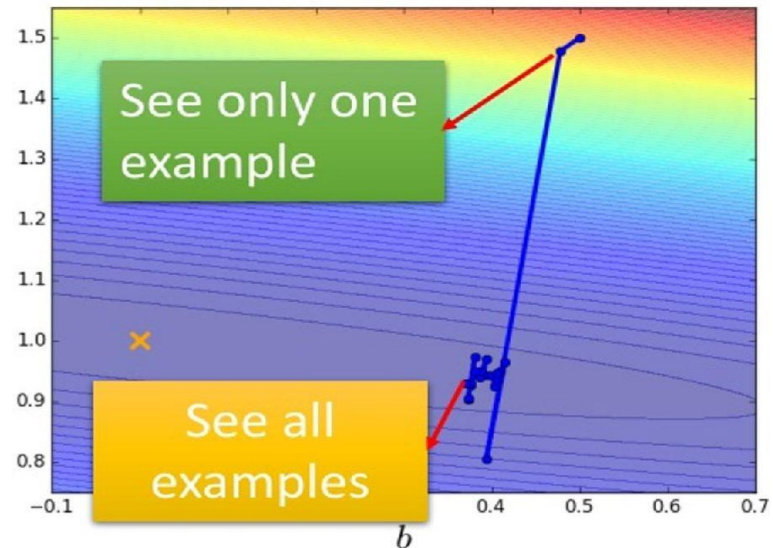
Gradient Descent

Update after seeing all examples



Stochastic Gradient Descent

Update for each example
If there are 20 examples,
20 times faster.



Difference Between Gradient Descent (GD) and Stochastic Gradient Descent (SGD)

Gradient Descent (GD)

Uses the **entire training dataset** to update weights.

Weight update happens **once per epoch**.

Training is **slower**.

Requires **more memory**.

Produces a **smooth optimization path**.

Suitable for **small datasets**.

More **stable** convergence.

Less chance of oscillation.

Takes **more computation time** per update.

Can get stuck in **local minima**.

Stochastic Gradient Descent (SGD)

Uses **one training example** to update weights.

Weight update happens **after every training sample**.

Training is **faster**.

Requires **less memory**.

Produces a **noisy optimization path**.

Suitable for **large datasets**.

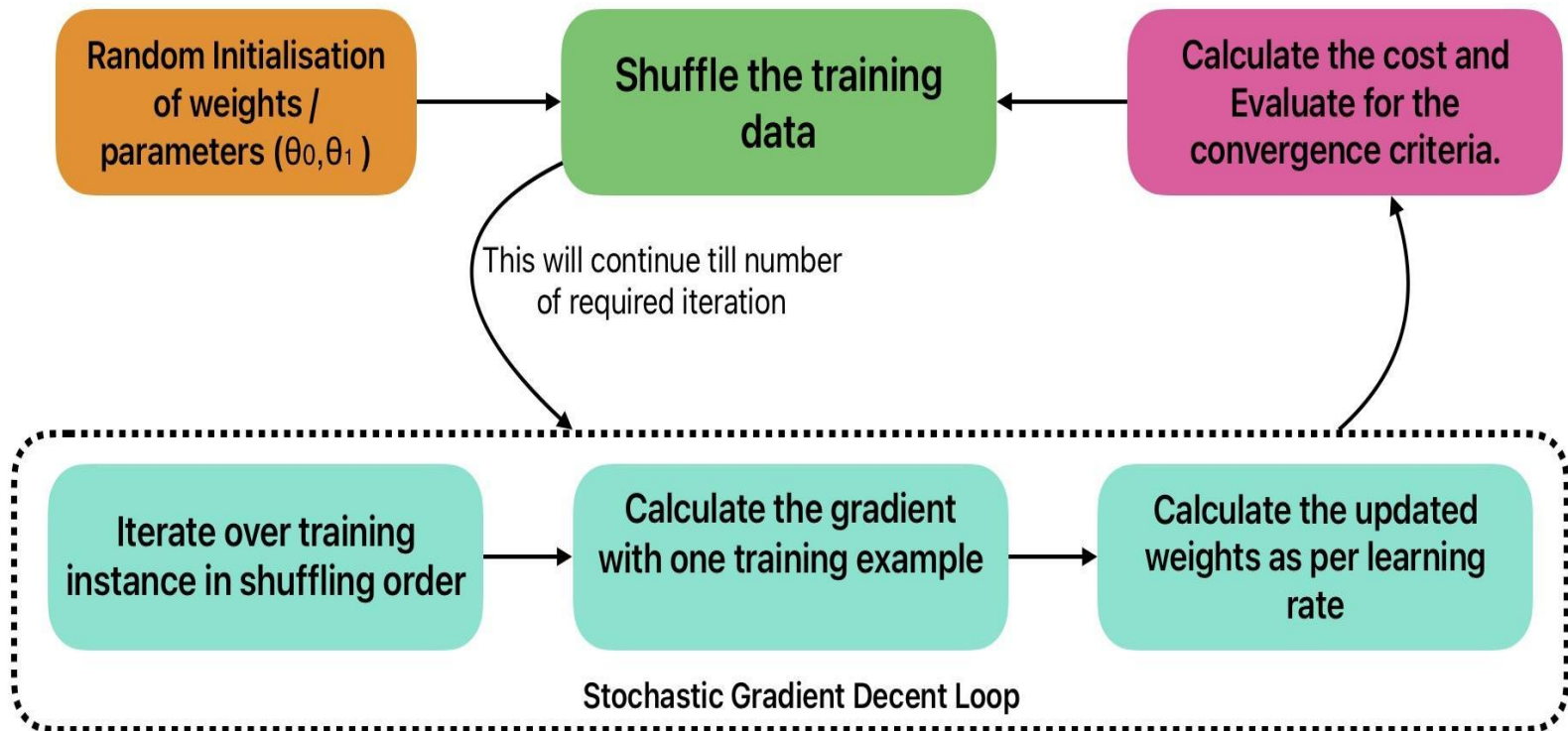
Less stable due to random updates.

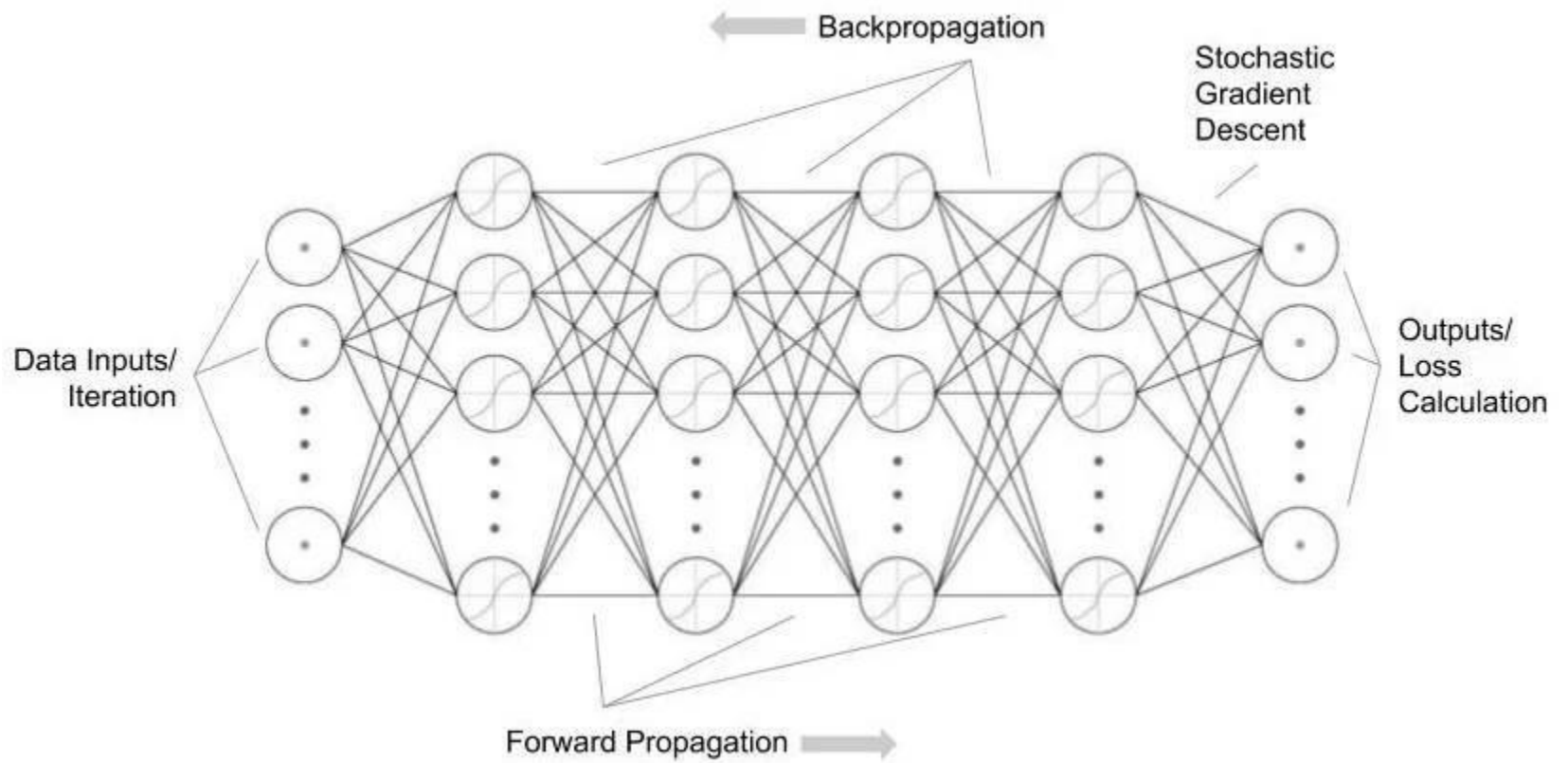
May oscillate around the minimum.

Takes **less computation time** per update.

Has a better chance of escaping **local minima** due to randomness.

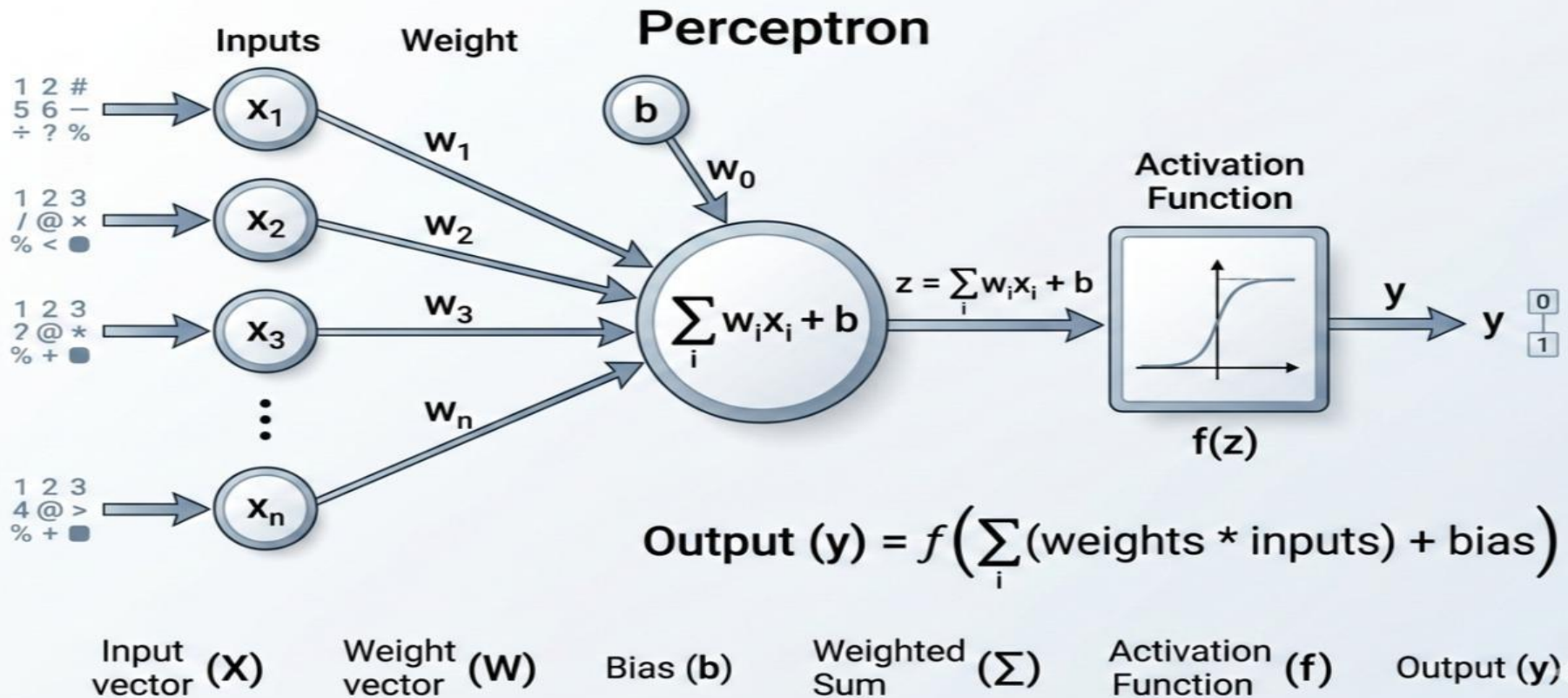
Steps in Stochastic Gradient Decent





Perceptron

- Single-layer neural network for binary classification.



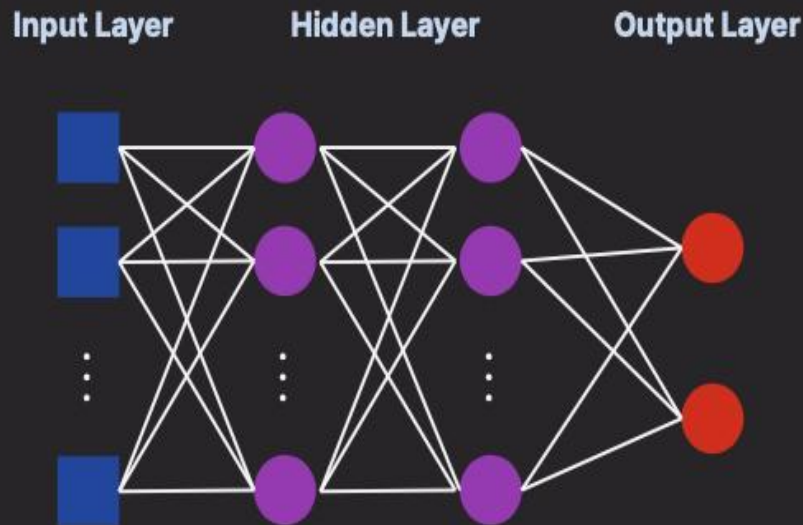
Multilayer Perceptron

- Neural network with one or more hidden layers.
- **Definition**
- A **Multilayer Perceptron (MLP)** is a type of **Artificial Neural Network (ANN)** that consists of **one input layer, one or more hidden layers, and one output layer**. It uses **backpropagation** for learning and can solve **complex, non-linear problems** that a single-layer perceptron cannot.

Cont..

MLP (Multilayer Perceptron)

- A basic form of neural network that consists of multiple layers of nodes.
- Each node is connected to every node in the previous and the next layers.



Working of MLP (Step by Step)

- Example: College Placement Prediction
- Suppose a student has:

Feature	Value
CGPA	8.8
Aptitude	85
Coding	92

Step 1: Input Layer

- The student details are fed into the network.
- CGPA = 8.8
- Aptitude = 85
- Coding = 92
- 
- Input Layer

Step 2: Hidden Layer

- The processed information reaches the output layer, which predicts:
- Output = 0.96
- Meaning:
- **96% probability of getting placed.**

Advantages of MLP

-  Solves **non-linear** problems.
-  Learns complex patterns from data.
-  High prediction accuracy.
-  Can handle multiple input features.
-  Used in many AI and Deep Learning applications.

Disadvantages of MLP

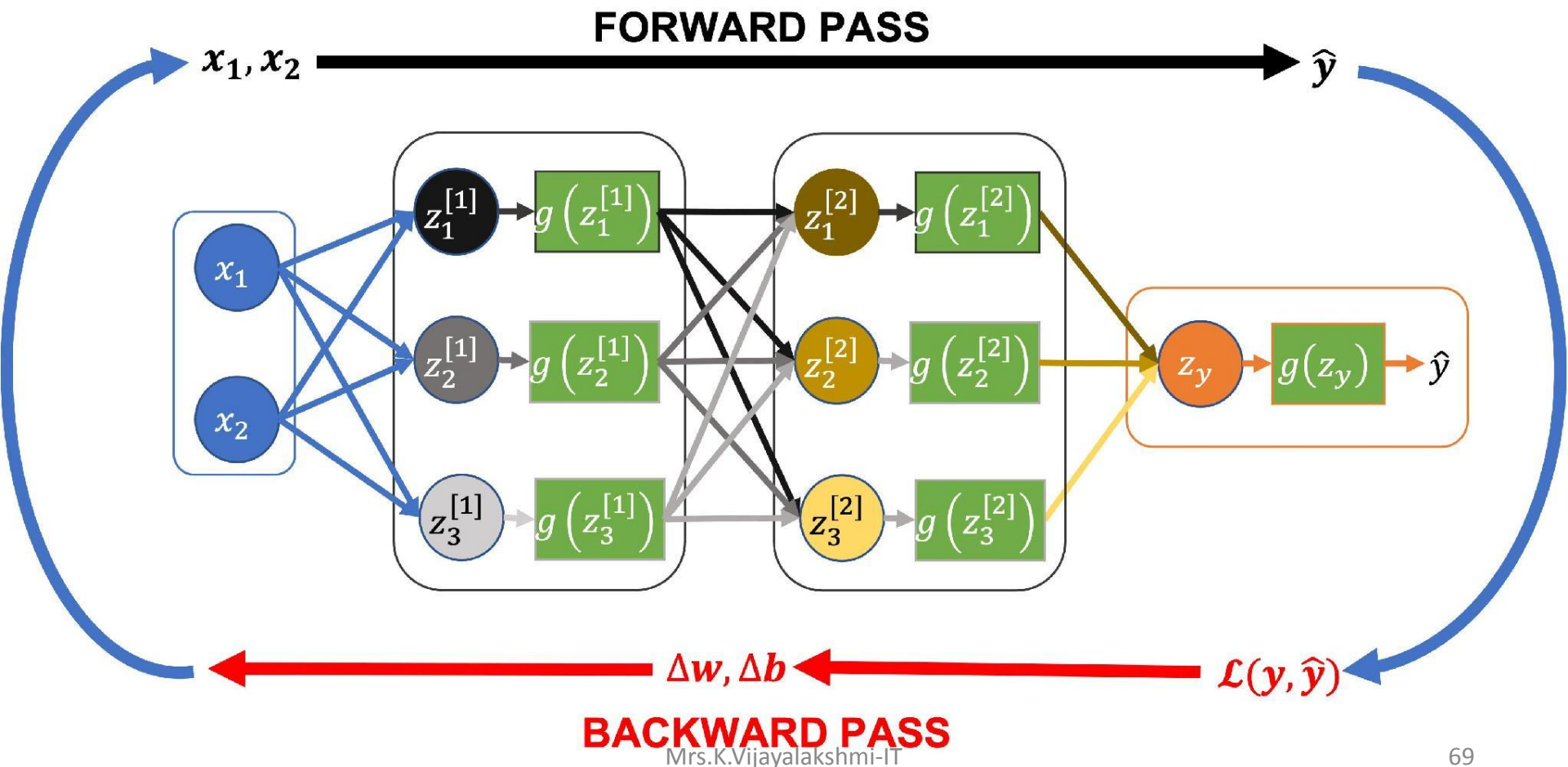
-  Requires a large amount of training data.
-  Training can be slow.
-  Needs more computational resources.
-  Can overfit if not properly regularized.
-  Difficult to interpret (black-box nature).

Applications

- Image Classification
- Handwritten Digit Recognition
- Speech Recognition
- Medical Diagnosis
- Stock Market Prediction
- College Placement Prediction
- Fraud Detection

Backpropagation

- Propagates error backward to update weights.



Advantages & Disadvantages

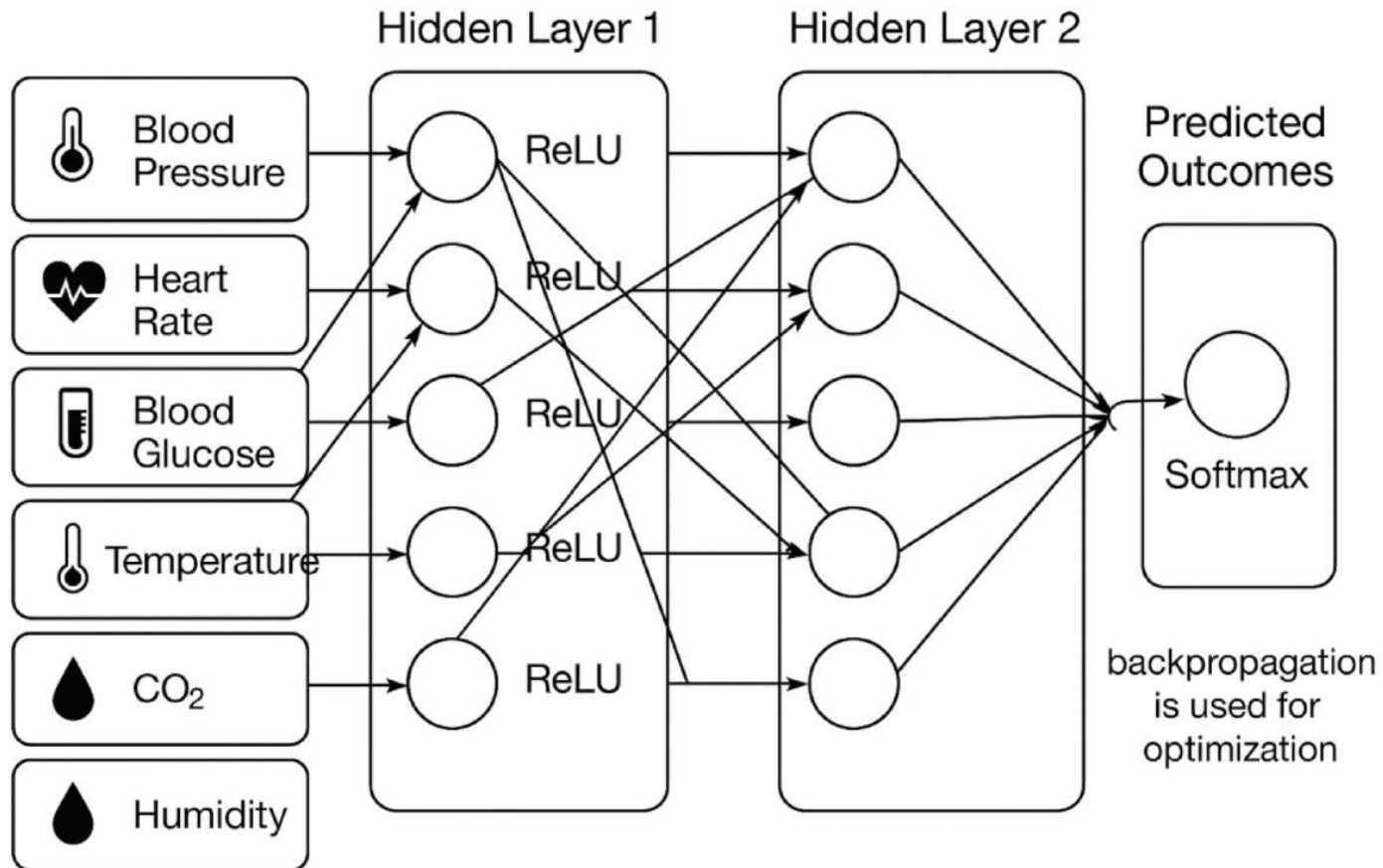
- High accuracy but requires large datasets and computation.

Applications in Real World

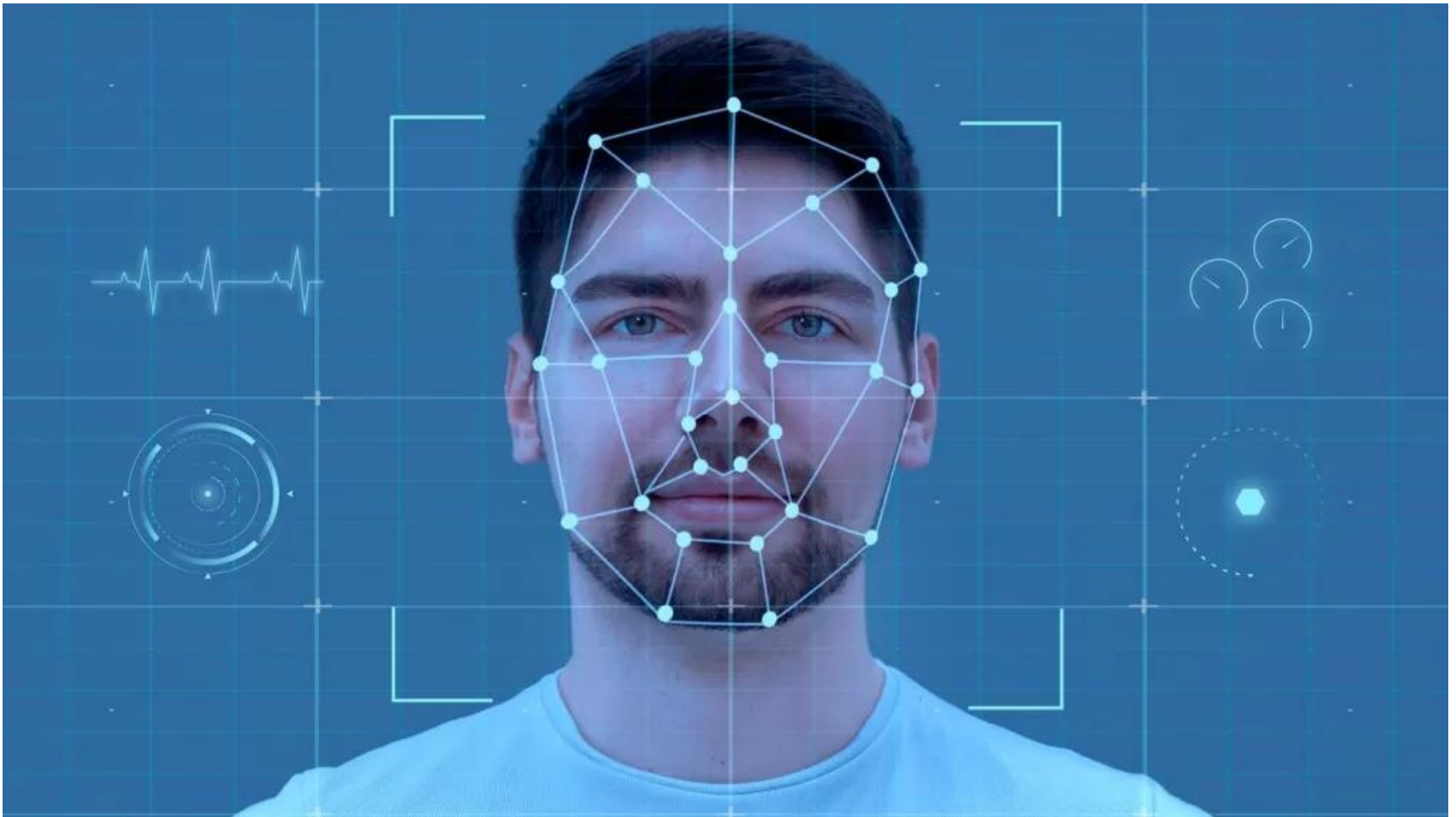
- Healthcare, Banking, Transportation, Security, Robotics.



Healthcare

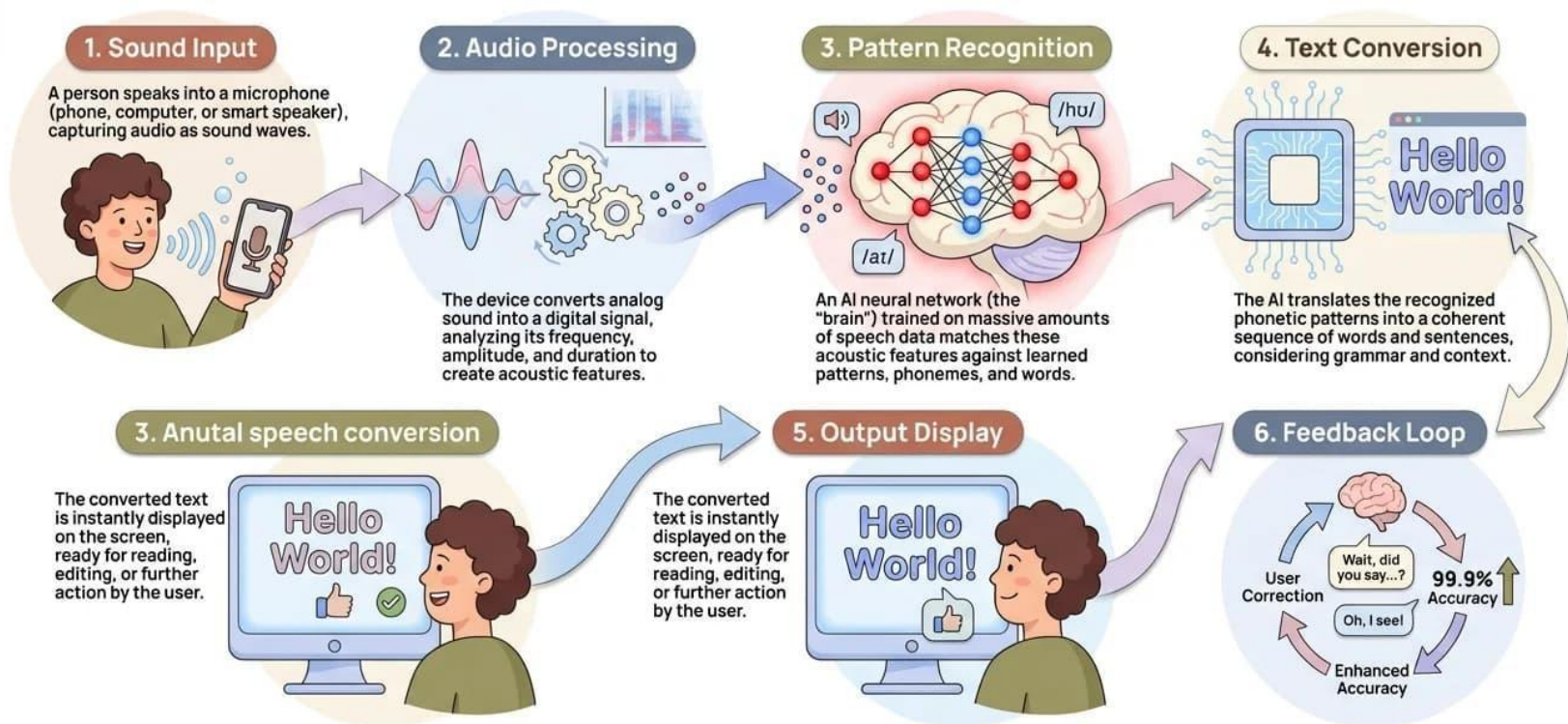


Security

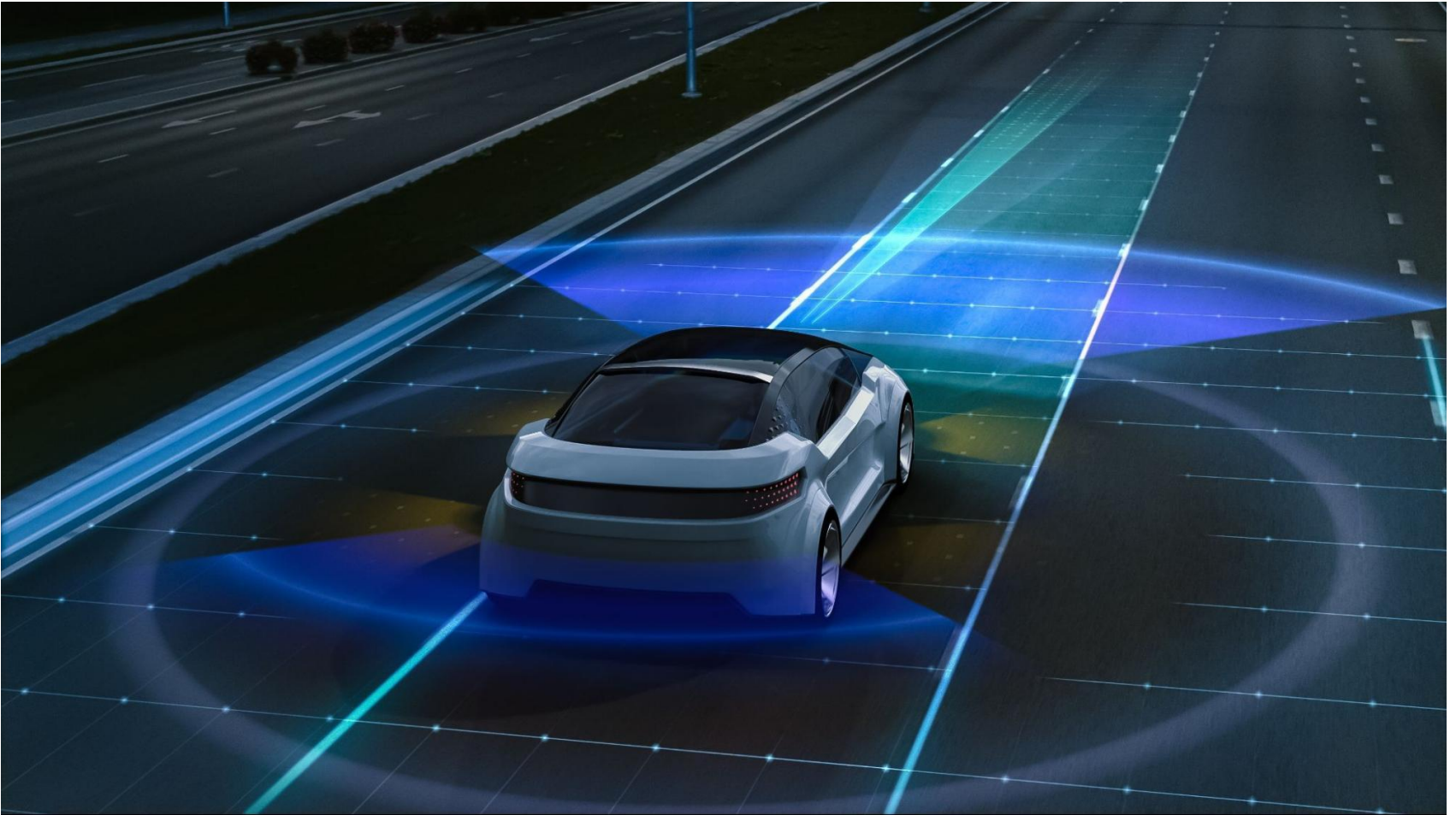


Applications in Real World

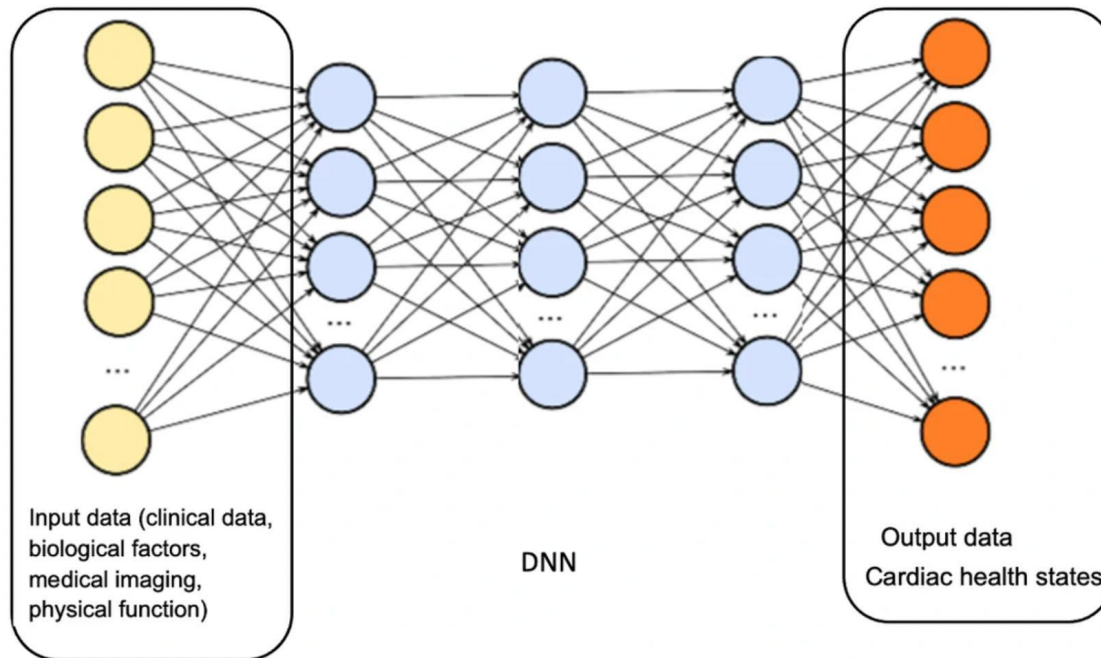
How AI Speech Recognition Works



Cont..



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Multimodal Data Analysis and Interpretation



Conclusion

- Neural Networks are powerful tools for modern AI systems.

1.Which of the following introduces non-linearity into ANN?

- A) Weight
- B) Bias
- C) Activation Function
- D) Input Layer

2.Which activation function is mainly used for binary classification?

- A) Softmax
- B) Sigmoid
- C) ReLU
- D) Linear

3.The output range of the Sigmoid function is

- A) -1 to 1
- B) 0 to 1
- C) 0 to ∞
- D) $-\infty$ to ∞

4. Which activation function is most widely used in Deep Learning?

- A) Tanh
- B) ReLU.
- C) Sigmoid
- D) Softmax

5. The output range of the Tanh function is

- A) 0 to 1
- B) -1 to 1 .
- C) 0 to ∞
- D) $-\infty$ to ∞

6.Which activation function is used for multi-class classification?

- A) Sigmoid
- B) ReLU
- C) Softmax
- D) Tanh

7.The main purpose of weights in ANN is to

- A) Store data
- B) Increase importance of inputs
- C) Reduce neurons
- D) Remove errors

- **8. Bias helps to**

- A) Shift the activation function.
- B) Remove neurons
- C) Store weights
- D) Increase inputs

9.Gradient Descent is used to

- A) Increase error
- B) Minimize loss.
- C) Delete weights
- D) Increase neurons

10.Gradient Descent is an

- A) Activation Function
- B) Optimization Algorithm
- C) Programming Language
- D) Dataset

11.Gradient Descent updates

- A) Inputs
- B) Outputs
- C) Weights and Biases
- D) Dataset

12.Which algorithm updates weights after processing the entire dataset?

- A) SGD
- B) Gradient Descent
- C) Adam
- D) RMSPProp

13.A Perceptron is a

- A) Multi-layer Network
- B) Single-layer Neural Network
- C) CNN
- D) RNN

14.Perceptron is mainly used for

- A) Image Processing
- B) Binary Classification
- C) Speech Recognition
- D) Object Detection

15.A Perceptron can solve

- A) Linear Problems
- B) Non-linear Problems
- C) Image Problems
- D) Video Problems

16.MLP stands for

- A) Multiple Learning Process
- B) Multilayer Perceptron
- C) Machine Learning Process
- D) Multi Logical Program

17. MLP contains

- A) Only Input Layer
- B) Only Hidden Layer
- C) One or More Hidden Layers
- D) No Hidden Layer

18. MLP is trained using

- A) Bubble Sort
- B) Backpropagation
- C) DFS
- D) BFS

19. MLP can solve

- A) Linear Problems
- B) Non-linear Problems
- C) Arithmetic Only
- D) Sorting Problems

20.Backpropagation is used for

- A) Image Compression
- B) Weight Updating
- C) Encryption
- D) Searching

21.Backpropagation is mainly used for:

- A) Data compression
- B) Training neural networks
- C) Image resizing
- D) Data encryption

22.Backpropagation is commonly used in:

- A) Single-layer Perceptron
- B) Multilayer Perceptron (MLP)
- C) Decision Tree
- D) K-Means.

23.Backpropagation updates the:

- A) Input data
- B) Weights and biases
- C) Output labels
- D) Dataset size

24. Backpropagation calculates:

- A) Mean value
- B) Error gradient.
- C) Dataset size
- D) Accuracy only

25. Backpropagation uses the _____ rule.

- A) Bayes
- B) Chain
- C) Newton
- D) Hebbian

26. The main objective of Backpropagation is to:

- A) Increase error
- B) Minimize the loss function
- C) Increase dataset size
- D) Remove neurons

27. Which activation function is suitable for Backpropagation?

- A) Step Function
- B) Differentiable Activation Function
- C) Random Function
- D) Identity Only

28.Backpropagation helps improve:

- A) Network accuracy
- B) Image quality
- C) File size
- D) Memory capacity

29.Backpropagation is a:

- A) Supervised learning algorithm
- B) Unsupervised learning algorithm
- C) Reinforcement learning algorithm
- D) Clustering algorithm
- **30. The purpose of Backpropagation is to:**
 - A) Increase loss
 - B) Reduce prediction error
 - C) Increase computation time
 - D) Remove hidden layers

Working of Neural Networks (10 Questions)

1. The process of passing inputs through the network is called .
2. The output of a neuron is calculated using an .
3. The weighted sum is calculated before applying the .
4. Training a neural network requires a .
5. The difference between predicted and actual output is called .
6. The goal of training is to minimize the .
7. The learning process adjusts the .
8. A neural network learns by updating .
9. The output of one layer becomes the of the next layer.
10. Neural networks learn through repeated .

Gradient Descent (10 Questions)

1. Gradient Descent is an _____ algorithm.
2. Gradient Descent minimizes the _____.
3. The step size in Gradient Descent is called the _____.
4. Learning rate is represented by _____.
5. Too high a learning rate may cause **divergence**.
6. Too low a learning rate results in **slow convergence**.
7. Gradient Descent updates **weights** after calculating gradients.
8. The gradient indicates the direction of **maximum increase**.
9. Gradient Descent moves in the **opposite** direction of the gradient.
10. The objective is to reach the _____.

Stochastic Gradient Descent (SGD)

(Questions)

1. SGD stands for .
2. SGD updates weights after processing .
3. SGD is generally than Batch Gradient Descent.
4. SGD introduces during training.
5. SGD requires less .
6. SGD is suitable for .
7. SGD may produce a optimization path.
8. Mini-batch Gradient Descent uses a of samples.
9. SGD helps the model escape .
10. SGD performs frequent .

Perceptron (Questions)

1. A Perceptron is a _____ neural network.
2. A Perceptron is used for _____ .
3. The Perceptron uses _____ and bias.
4. The output of a Perceptron is usually _____ .
5. The Perceptron learning rule updates _____ .
6. A Perceptron cannot solve the _____ problem.
7. The decision boundary of a Perceptron is _____ .
8. A Perceptron works only for _____ data.
9. The activation function commonly used is the _____ .

Backpropagation (Questions)

1. Backpropagation is used to train **multilayer neural networks**.
2. Backpropagation calculates the **error gradient**.
3. Errors are propagated from **output to input**.
4. Backpropagation uses the **chain rule** of calculus.
5. The objective is to reduce the **loss**.
6. Weights are updated after computing **gradients**.
7. Backpropagation requires a **differentiable** activation function.
8. The algorithm repeatedly adjusts **weights and biases**.
9. Backpropagation improves model **accuracy**.
10. Backpropagation works together with **Gradient Descent**.